Week 5 — Measures and first dashboard

Group 4b · Ride-hailing trips · your week 5 deliverable

Group	group4b
Phase lead this week	
Who wrote the measures	
Who built the dashboard	
Date completed	

Your question

Which pickup zones generate the most revenue, and where are cancellations concentrated?

Ugly is expected this week

You are proving the numbers are right, not designing anything. Week 6 is when it gets to look good. A beautiful dashboard built on a wrong total is worse than an ugly one built on a right total, because nobody questions it.

How to read the numbers in this document

Every figure below was computed independently from the cleaned trips table (28,000 rows, week 3 decisions applied), replicating each measure's logic exactly. The "Power BI says" column therefore holds the value each measure must return — read your own card in Power BI and confirm it matches before you submit. Items that need a person at the screen (the slicer test, the screenshot) are marked "confirm live."

1. Your measures

Five is plenty. Write the DAX for each one here — we will be reading it, so format it readably. Column names below are the underlying names in group4b.trips_clean; if you renamed a column in Model view (e.g. fare → Fare), use the renamed name inside the brackets.

1. Total Fares

The DAX:

```
Total Fares =
CALCULATE (
    SUM ( trips_clean[fare] ),
    trips_clean[status] = "Completed"
)

-- Revenue only: fares on completed trips. SUM skips the blank
-- fares left by cleaning, so nothing is invented or zero-filled.
```

2. Trips

The DAX:

```
Trips =
COUNTROWS ( trips_clean )

-- Every trip attempt (our grain), completed or cancelled.
```

3. Completion Rate

The DAX:

```
Completion Rate =
VAR CompletedTrips =
    CALCULATE (
        COUNTROWS ( trips_clean ),
        trips_clean[status] = "Completed"
    )
RETURN
    DIVIDE ( CompletedTrips, [Trips] )

-- Format as percentage, 1 decimal. DIVIDE returns blank, not an
-- error, if a slicer selection leaves zero trips.
```

4. Average Fare

The DAX:

```
Average Fare =
CALCULATE (
    AVERAGE ( trips_clean[fare] ),
    trips_clean[status] = "Completed"
)

-- AVERAGE ignores blank fares, so this is the mean of completed
-- trips that HAVE a fare. Do not write [Total Fares] / completed
-- trips: that would divide by trips with no fare and understate it.
```

5. Cancellations

The DAX:

```
Cancellations =
CALCULATE (
    COUNTROWS ( trips_clean ),
    trips_clean[status] <> "Completed"
)

-- Both cancelled statuses (by rider, by driver). Written on
-- status, so it needs no helper column in the model.
```

2. Proving the numbers are right

Every measure, checked against a SQL query in DBeaver. This is the most important section in the form — a measure that returns a plausible wrong number is worse than one that errors, because nobody notices.

```
-- the pattern, in DBeaver
SELECT round(sum(fare), 2) FROM group4b.trips_clean;
-- then drop the measure into a card in Power BI
-- the two must match exactly
```

The SQL used for each measure:

```
-- Total Fares
SELECT round(sum(fare), 2) FROM group4b.trips_clean WHERE status = 'Completed';
-- Trips
SELECT count(*) FROM group4b.trips_clean;
-- Completion Rate
SELECT round(100.0 * count(*) FILTER (WHERE status = 'Completed') / count(*), 2)
FROM group4b.trips_clean;
-- Average Fare  (avg() skips NULLs, exactly like DAX AVERAGE)
SELECT round(avg(fare), 2) FROM group4b.trips_clean WHERE status = 'Completed';
-- Cancellations
SELECT count(*) FROM group4b.trips_clean WHERE status <> 'Completed';
```

Measure name	Power BI says	SQL says	Match? If not, why
Total Fares	GH₵561,454.89	561454.89	Yes. Deliberately not the GH₵655,163.97 from the week 4 plumbing check: that unfiltered sum includes GH₵93,709.08 of fares sitting on cancelled trips (see section 3).
Trips	28,000	28000	Yes. Also ties to the week 3 row count (28,112 − 112 duplicates).
Completion Rate	85.8%	85.83	Yes. 24,033 completed ÷ 28,000 = 85.83%. Compare at the same rounding — the card shows 85.8% at one decimal.
Average Fare	GH₵24.64	24.64	Yes. Mean of the 22,783 completed trips that have a fare. The tempting shortcut, Total Fares ÷ 24,033, gives 23.36 and would not match.
Cancellations	3,967	3967	Yes. 2,547 by rider + 1,420 by driver; Trips − completed = 28,000 − 24,033 = 3,967.

Two cross-checks that do not depend on SQL: Completion Rate + cancellation share must equal 100% (85.83% + 14.17%), and Total Fares by zone_name must add back to GH₵561,454.89 with no blank zone row — that one proves the pickup-zone relationship.

If any did not match — what was wrong, and what did you change? (Usually the model, not the measure.)

None are expected to mismatch, but this is where to look if one does. Total Fares off by exactly GH₵93,709.08 means the status filter is missing (the unfiltered week 4 total is back). Average Fare reading 23.36 means it was written as Total Fares ÷ completed trips instead of AVERAGE. A total that is short by a round number of trips means a relationship is silently dropping rows — check for a (Blank) zone or date row first, then check that dim_date still covers 2024-01-01 to 2025-12-30. Fix the model before touching the DAX.

3. Assumptions you had to make

Some measures need a decision that the data cannot settle, exactly like dayfirst in week 3. State them plainly — you will be asked to defend them in week 8.

Does a cancelled trip count in your average fare? Should it?

Our decision, and why:
No — cancelled trips are excluded from Average Fare, and from Total Fares for the same reason. An average fare should describe what a completed ride earns, and a cancelled ride earned nothing. The data makes this a real decision, not a formality: cancelled trips are not fare-less. 3,779 of the 3,967 cancelled trips carry a fare, worth GH₵93,709.08 in total, averaging GH₵24.80 — almost identical to the GH₵24.64 average on completed trips, and spread the same way (median about GH₵22 in both). Rider and driver cancellations look alike (GH₵24.84 vs GH₵24.72 average). That pattern reads like a quoted fare recorded at booking, not money collected; a cancellation fee would not be charged on driver cancellations and would not mirror the completed-trip distribution. The choice barely moves the average (24.67 with them, 24.64 without) but moves Total Fares by 17% (655,163.97 vs 561,454.89), which is exactly why it has to be written down. To be confirmed with the course team on the weekly call.

Any other assumptions buried in your measures:
1. Completed is the only revenue-earning status. Two statuses exist for cancellation, and both are treated identically in Cancellations. 2. Completed trips with no fare (1,250 of 24,033, 5.2%, left as NULL in week 3) are excluded from Total Fares and Average Fare. Total Fares is therefore a floor, not an exact revenue figure. Average Fare uses 22,783 trips, not 24,033. 3. Trips counts every attempt including cancellations (our grain), so it is not a count of rides taken. Completion Rate uses all 28,000 as its denominator and, with only three statuses in the data, equals 100% minus the cancellation share. 4. Every zone breakdown uses the pickup zone (the active relationship). Dropoff-based views need USERELATIONSHIP and are not in this draft. 5. The 70 trips mapped to “Unknown Driver” stay in every measure — none of the five depends on the driver. Currency is assumed to be GH₵ throughout.

4. Your dashboard page

One page. Cards along the top, a trend, a breakdown, at least one slicer.

Which measures are on the cards?
Five cards in one row: Total Fares, Trips, Completion Rate, Average Fare, Cancellations — one per measure, no filters applied to the cards themselves so they respond only to the slicers.

What is the trend chart showing, and is it using dim_date rather than the fact table's own date column?
A column chart of Total Fares by month across all 24 months (Jan 2024 – Dec 2025). Yes, it uses dim_date: the axis is dim_date[Year] and dim_date[Month], with Month sorted by MonthNumber so it reads Jan→Dec rather

than alphabetically. trips_clean[trip_date] is not used on any visual — using it would bypass the date table and the “Mark as date table” work from week 4.

What is the breakdown chart showing?

Two bar charts side by side, both by zones_clean[zone_name] (pickup zone). Left: Total Fares, sorted highest to lowest — answers “which pickup zones generate the most revenue.” Right: Completion Rate, sorted lowest first, so the zones with the most cancellations sit at the top — answers “where are cancellations concentrated.” A rate is used instead of the Cancellations count because a count mostly reflects how busy a zone is.

Which slicers did you add? Did you click them and watch every visual change?

Three slicers: Year (dim_date[Year]), Zone Type (zones_clean[zone_class]) and Vehicle Type (drivers_clean[vehicle_type]).

Confirm live: click each one and check that all five cards, the trend and both bar charts change. Zone Type is the useful test — it filters through the pickup-zone relationship, so selecting Suburban should leave 7 zones on both bar charts. Selecting 2025 should leave 12 months on the trend. If a visual does not move, its table is not connected to that slicer in Model view.

Paste a screenshot of the page below

[Paste your own screenshot of the finished page here before submitting — it can only come from your .pbix.]

5. What does it actually say?

You now have real numbers for the first time. What do they tell you about your question?

The single most interesting thing you can see so far:

Nothing is concentrated. Revenue is close across all 15 pickup zones: Teshie leads at GH₵39,577 and Adenta trails at GH₵35,636, an 11% gap, and the top three zones hold 21% of revenue against the 20% an even split would give. Cancellations are just as flat: completion rate runs from 86.9% (Airport City) down to 84.7% (Osu), against 85.8% overall. Osu, Dansoman and Madina are the weakest, Airport City, Lapaz and East Legon the strongest, but a 2-point range is a nudge, not a hotspot. For the operations lead, that means no zone is a clear place to add or pull drivers on these two measures alone.

Anything that surprised you, or contradicted what you expected in week 1:

Busiest is not the same as most valuable. The week 1 query ranked zones by trip count (with duplicates still in); revenue reorders them. Madina is the 4th busiest zone but only 7th for revenue, because it completes just 85.0% of its trips; Airport City is 5th busiest but 3rd for revenue, because it completes 86.9%. Volume alone would have pointed drivers at the wrong zones.

The second surprise was that cancelled trips carry fares at all — GH₵93,709.08 of them (section 3). Week 1 only looked at fare as a single column and would not have shown it.

Anything that looks wrong rather than surprising — and what you plan to check:

1. Fares on cancelled trips. Check whether the amounts match a quote (compare against distance), and ask the course team whether cancelled-trip fares are quotes or fees. Driver cancellations carry them too (1,357 of 1,420), which argues against a fee.

2. Missing fares are not spread evenly: they range from 68 to 102 completed trips per zone. Filling each zone's gaps at its own average fare leaves the top five zones in the same order but reshuffles the middle (Haatso would jump from 14th to 11th). So the top of the ranking is reliable, the middle is not. Plan: report ranks as groups, not exact positions, until the missing fares are explained.

3. The monthly trend moves a lot for a flat network (best month Aug 2024, GH₵25,098; weakest Sep 2025, GH₵21,658; cancellation rate peaking at 16.2% in Jun 2025). Check whether this is seasonality or just fewer days and trips in a month before reading anything into it.

6. Ready for week 6?

Week 6 is iteration — fixing what this draft exposed and making it readable. Think about what it exposed.

What is missing from the model that you now wish you had? (A dimension, a column, a derived field.)

1. A Cancellation Rate measure (and a Cancelled By Rider / By Driver split) — the breakdown currently leans on Completion Rate, which reads backwards for a question about cancellations.

2. A derived Fare Status column (Has fare / Missing fare) so the 1,250 fareless completed trips can be counted and filtered rather than silently skipped by SUM.

3. A time-of-day column. trip_date is a date only, so nothing here can say when cancellations happen.

4. A friendlier status dimension, so Completed / Cancelled By Rider / Cancelled By Driver can sit on a slicer with a fixed sort order.

Which of your week 1 sub-questions can this dashboard already answer?

Which pickup zones generate the most revenue (Total Fares by zone). Where cancellations are concentrated by zone, by zone type and over time (Completion Rate and Cancellations with the slicers). How big the trip base is and what a typical fare is (Trips, Average Fare). Who cancels more: Cancellations split by status shows riders account for 64% (2,547 vs 1,420).

Which cannot be answered yet, and what would it take?

Whether cancellations cluster at particular times of day: needs a timestamp, the data holds dates only. Why trips are cancelled: needs a cancellation reason field. Whether a zone is under-supplied with drivers: needs driver availability or online-hours data. True revenue per zone: needs the 1,250 missing fares explained or recovered, and an answer on whether cancelled-trip fares are real money.

7. Questions for the weekly call

Are the fares recorded on cancelled trips (GH₵93,709.08 across 3,779 trips) quoted fares or cancellation fees? It decides whether “revenue” is GH₵561k or closer to GH₵655k.

Is there any way to recover or explain the 1,250 completed trips with no fare, given they are 5.2% of completed trips and enough to reorder the middle of the zone ranking?

Note on week 4: that form listed is_cancelled and cancelled_by as columns in trips_clean, but the week 3 notebook's trips_clean does not contain them. The measures above are written on status for that reason. Should those two columns be added to the cleaned table, or dropped from the week 4 notes?

Before you submit this

Five measures written, named in plain English and formatted. Every one checked against SQL. One dashboard page with cards, a trend, a breakdown and a working slicer. The .pbix committed to dashboard/. Measure definitions can be discussed openly with group4a. Your dashboard and your findings stay yours.